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Privacy-Preserving Student Attendance System Based on Face Recognition with Encrypted Data Embedding in Images Using Steganography

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ABSTRACT

This thesis presents the design and implementation of a secure and privacy-preserving student attendance system based on face recognition. The proposed system integrates deep learning-based facial recognition, symmetric encryption, and image steganography to provide a robust, real-time solution that ensures both operational efficiency and data confidentiality.

Face recognition was implemented using a deep embedding approach trained on the VGGFace2 dataset, achieving a recognition accuracy of 93.8%, precision of 92.9%, recall of 92.2%, and an F1-score of 91.0%. Attendance data is encrypted using AES encryption and embedded into captured images using Least Significant Bit (LSB) steganography, resulting in minimal perceptible distortion (PSNR: 41.2 dB, SSIM: 0.988). System evaluation demonstrates near real-time processing capability, high classification reliability, and robust protection of sensitive information against unauthorized access.

The integration of biometric recognition with cryptography and steganography establishes a multi-layered security framework for attendance management systems, addressing existing challenges in data privacy, system integrity, and operational efficiency. The proposed approach offers a practical and scalable solution for institutions seeking secure, automated, and privacy-conscious attendance tracking.

Keywords: Face recognition, privacy-preserving, encryption, steganography, attendance system, VGGFace2, deep learning, secure data embedding.

ABSTRACT IN ARABIC

تهدف هذه الرسالة إلى تصميم وتنفيذ نظام حضور آمن يحافظ على خصوصية الطلاب ويعتمد على التعرف على الوجه. يدمج النظام المقترح بين التعرف على الوجه باستخدام التعلم العميق، والتشفير المتماثل، وتقنيات إخفاء البيانات داخل الصور (Steganography)لتقديم حل متكامل يعمل في الوقت الفعلي ويضمن كل من الكفاءة التشغيلية وسرية البيانات.

تم تنفيذ التعرف على الوجه باستخدام نموذج embedding عميق مدرب على مجموعة بياناتVGGFace2 ، محققًا دقة في التعرف تصل إلى 93.8%، ودقة إيجابية92.9% (Precision) ، واسترجاع92.2% (Recall) ، و .91.0% F1-scoreتُشفر بيانات الحضور باستخدام تشفير AES وتدمج ضمن الصور الملتقطة باستخدام تقنية(Least Significant Bit (LSB ، مع الحفاظ على جودة الصورة وعدم وجود تشويه ملحوظ .(PSNR: 41.2 dB, SSIM: 0.988) تُظهر نتائج تقييم النظام قدرة معالجة شبه فورية، ومصداقية عالية في التصنيف، وحماية قوية للمعلومات الحساسة ضد الوصول غير المصرح به.

يمثل دمج التعرف على الوجه مع التشفير وتقنيات إخفاء البيانات إطارًا أمنيًا متعدد الطبقات لأنظمة إدارة الحضور، معالجًا التحديات القائمة في حماية الخصوصية وسلامة النظام والكفاءة التشغيلية. يقدم النظام المقترح حلًا عمليًا وقابلًا للتوسع للمؤسسات التي تسعى إلى تتبع حضور آمن، آلي، ويولي اهتمامًا كبيرًا للخصوصية.

الكلمات المفتاحية :التعرف على الوجه، الحفاظ على الخصوصية، التشفير، إخفاء البيانات داخل الصور، نظام الحضور، VGGFace2، التعلم العميق، تضمين البيانات الآمن.

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LIST OF ABBREVIATIONS

ANN	Artificial Neural Network
CNN	Convolutional Neural Network
DCT	Discrete Cosine Transform
DFD	Data Flow Diagram
ERD	Entity Relationship Diagram
GUI	Graphical User Interface
KNN	K-Nearest Neighbours
LBP	Local Binary Patterns
ML	Machine Learning
PCA	Principal Component Analysis
RFID	Radio Frequency Identification
SVM	Support Vector Machine

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Chapter 1 Introduction

1.1. Overview

In recent years, biometric technologies have become a cornerstone in modern intelligent systems, particularly in identity verification and access control applications. Among these technologies, face recognition has emerged as one of the most widely adopted methods due to its non-intrusive nature, ease of deployment, and compatibility with existing camera infrastructures. This has led to its rapid integration into attendance management systems across educational institutions and organizations.

Traditional attendance systems, including manual roll-calling, RFID cards, and fingerprint-based systems, suffer from several limitations such as time inefficiency, susceptibility to proxy attendance, hygiene concerns, and operational overhead. Consequently, automated attendance systems based on face recognition have been proposed to address these challenges by enabling real-time identification and seamless attendance logging [1].

Recent advancements in deep learning, particularly convolutional neural networks (CNNs), have significantly improved the accuracy and robustness of face recognition systems. Transfer learning approaches using pre-trained deep neural networks have demonstrated high accuracy in real-world attendance scenarios while maintaining reasonable computational efficiency [2].

Despite these advancements, the widespread deployment of face recognition systems raises critical concerns regarding data privacy and security. The storage of facial images and associated identity information in centralized databases introduces significant risks, including data breaches, unauthorized access, and misuse of sensitive biometric data. A systematic review of face recognition-based attendance systems highlights that security and privacy concerns are among the most critical challenges in their implementation, with a significant proportion of studies emphasizing the importance of secure data handling mechanisms [3].

To address these concerns, privacy-preserving techniques have gained increasing attention. Research demonstrates that it is possible to maintain high recognition performance while protecting sensitive visual information through techniques such as data obfuscation and feature transformation [4]. Furthermore, cryptographic methods can be employed to secure attendance data, while steganography provides an additional layer of protection by embedding encrypted information within digital images in a manner that is imperceptible to human observers.

This thesis proposes a novel privacy-presing student attendance system that integrates face recognition, encryption, and image steganography. The proposed approach eliminates the need

for explicit storage of sensitive data by embedding encrypted attendance information directly within captured images, thereby enhancing both security and privacy without compromising system performance.

1.2 Problem Definition

Despite the effectiveness of face recognition-based attendance systems, existing solutions predominantly rely on centralized storage of biometric data and attendance records. This architectural design introduces several critical vulnerabilities:

- Exposure of sensitive biometric data to unauthorized access
- Risk of large-scale data breaches
- Lack of privacy-preserving mechanisms in data storage
- Susceptibility to spoofing and identity fraud

Existing systems primarily focus on improving recognition accuracy while neglecting data protection. Even in systems that incorporate anti-spoofing mechanisms, the issue of secure data storage remains inadequately addressed [1].

Furthermore, conventional database-driven architectures store attendance logs separately from image data, making them more vulnerable to tampering and unauthorized modification. This separation also increases the attack surface, as both databases and image repositories must be secured independently.

Therefore, the core problem addressed in this research can be summarized as follows:

How can a face recognition-based attendance system be designed to ensure accurate identification while simultaneously preserving user privacy and protecting sensitive data from unauthorized access?

1.2. Research Questions

This thesis is guided by the following research questions:

1. How can face recognition be effectively integrated into a real-time attendance system with high accuracy and efficiency?
2. What are the primary privacy risks associated with storing biometric and attendance data in conventional systems?

3. How can cryptographic techniques be applied to secure attendance data without affecting system performance?
4. Can steganography be utilized to embed encrypted attendance data within images while maintaining image quality and imperceptibility?
5. What is the trade-off between privacy preservation and recognition accuracy in such systems?

1.3. Objectives and Contributions

The main objective of this thesis is to design and implement a secure and privacy-preserving student attendance system that leverages face recognition, encryption, and steganography.

The contributions of this thesis are summarized as follows:

- Develop a real-time face recognition-based attendance system using deep learning techniques
- Design a secure encryption mechanism to protect sensitive attendance data
- Implement a steganographic model to embed encrypted data within images
- Evaluate the system in terms of accuracy, security, robustness, and computational efficiency
- Analyze the trade-off between privacy preservation and system performance

1.4. Proposed Approach

The proposed system adopts a multi-layered architecture that integrates computer vision, deep learning, cryptography, and data hiding techniques to achieve secure and privacy-preserving attendance management.

1. System Workflow
   - Image Acquisition
     Real-time facial images are captured using a camera under unconstrained environmental conditions.
   - Face Detection and Alignment
     Advanced detection models (e.g., MTCNN or RetinaFace) are used to localize and normalize facial regions.

- Feature Extraction (Deep Embeddings)
  A deep convolutional neural network (e.g., FaceNet or ArcFace) is used to extract discriminative facial embeddings.
- Identity Matching
  Similarity metrics (e.g., cosine similarity) are applied to match extracted embeddings with stored templates.
- Attendance Generation
  Upon successful authentication, an attendance record is generated automatically.
- Data Encryption Layer
  The generated attendance data is encrypted using a strong symmetric encryption algorithm (e.g., AES-256).
- Steganographic Embedding Layer
  The encrypted data is embedded into a cover image using a robust steganographic technique (e.g., LSB-based or transform-domain methods).
- Secure Storage
  Instead of storing plaintext data or raw embeddings, only the stego-images are stored in the database.

2. Key Design Innovation

- Unlike conventional systems:

  No direct storage of raw sensitive data

  No reliance on single-layer protection
- Instead:

  Data is encrypted → then hidden → then stored

  Provides defense-in-depth security

1.5. Thesis Structure

- **Chapter 1: Introduction**

  Presents the research background, problem definition, research questions, objectives, and contributions.
- **Chapter 2: Literature Review**

  Provides a critical analysis of existing work in:

- Face recognition
- Privacy-preserving biometrics
- Cryptography
- Steganography

- **Chapter 3: Methodology**

  Describes the proposed system architecture, algorithms, and design rationale.

- **Chapter 4: Implementation**

  Details the system development, tools, frameworks, and integration process.

- **Chapter 5: Experimental Results and Evaluation**

  Evaluates system performance using quantitative and qualitative metrics.

- **Chapter 6: Conclusion and Future Work**

  Summarizes findings and outlines potential research extensions.

Chapter 2 Background and Related Work

2.1 Background

Face recognition has experienced a profound transformation over the past decades, evolving from classical appearance-based methods to highly sophisticated deep learning architectures. Early techniques such as Eigenfaces and Fisherfaces relied on linear subspace projections and statistical feature extraction. Although computationally efficient, these approaches were highly sensitive to variations in illumination, pose, and facial expressions, which limited their applicability in real-world environments.

The advent of deep learning, particularly convolutional neural networks (CNNs), marked a paradigm shift in face recognition. Deep models enabled automatic hierarchical feature extraction, allowing systems to learn highly discriminative representations directly from raw image data. This advancement led to a dramatic improvement in recognition accuracy, with modern systems achieving near-human or even superhuman performance under controlled conditions.

Despite these technological advancements, the deployment of face recognition systems in real-world applications has raised critical concerns related to privacy and security. Most existing systems rely on centralized architectures that store biometric data and identity information in databases. Such designs introduce significant risks, including data breaches, unauthorized access, and potential misuse of sensitive biometric information.

To address these concerns, privacy-preserving techniques have emerged as a crucial research direction. These include cryptographic methods, secure multi-party computation, and data obfuscation techniques. In parallel, steganography has been explored as a complementary approach for concealing sensitive information within digital media. When combined with encryption, steganography enables multi-layered protection by both securing and hiding the data.

This thesis builds upon these foundational domains by proposing an integrated framework that combines face recognition, encryption, and steganography into a unified, privacy-preserving attendance system.

2.2 Related Work

2.2.1 Deep Learning-Based Face Recognition

The introduction of deep learning has significantly advanced the state-of-the-art in face recognition. FaceNet [5] introduced a unified embedding space using a triplet loss function, enabling direct mapping of facial images into a Euclidean space where distances correspond to similarity. This approach improved scalability and achieved state-of-the-art performance on benchmark datasets. DeepFace [6] demonstrated that deep neural networks could achieve near-human performance in face verification by leveraging large-scale datasets and deep architectures. This work marked a major milestone in the field. ArcFace [7] further improved recognition performance by introducing an additive angular margin loss function, enhancing the discriminative power of learned features. This method achieved superior performance in large-scale recognition tasks.

In addition to model architectures, datasets such as VGGFace2 [8] have contributed significantly to improving generalization by providing diverse facial images across variations in pose, age, and lighting conditions. Despite these advancements, the primary focus of these approaches remains on maximizing accuracy, with limited consideration given to privacy and data protection.

2.2.2 Face Recognition-Based Attendance Systems

Face recognition has been increasingly adopted in attendance management systems due to its automation capabilities and ease of use. Karunakar et al. [9] proposed a smart attendance system based on transfer learning, demonstrating improved accuracy and reduced training time.

Earlier work by Wagh et al. [10] implemented a basic attendance system using traditional image processing techniques. While cost-effective, the system exhibited limited robustness and

accuracy. Arsenović et al. [11] conducted a comprehensive survey of face recognition-based attendance systems and identified key challenges, including environmental sensitivity, scalability issues, and lack of security mechanisms.

Although these systems demonstrate practical feasibility, they largely depend on centralized storage architectures, which expose them to significant security risks.

2.2.3 Privacy-Preserving Face Recognition

Privacy-preserving techniques have been proposed to mitigate risks associated with biometric data. Erkin et al. [12] introduced a framework based on secure multi-party computation, enabling face recognition without revealing sensitive data. While highly secure, the approach incurs significant computational overhead. Drozdowski et al. [13] analyzed privacy risks in face recognition systems, highlighting vulnerabilities such as data leakage, model inversion attacks, and unauthorized access. Yang et al. [14] explored face obfuscation techniques and demonstrated that it is possible to preserve recognition performance while reducing exposure of sensitive visual information. Despite these efforts, most approaches either sacrifice efficiency or lack integration with real-world systems.

2.2.4 Steganography and Data Hiding

Steganography provides an additional layer of security by concealing data within digital media. Baluja [15] introduced a deep learning-based method capable of hiding an entire image within another image. Zhu et al. [16] proposed HiDDeN, an end-to-end framework for robust data hiding that can withstand noise and transformations.Traditional methods such as LSB remain widely used due to their simplicity and efficiency, as discussed in [17] and [18]. Zhang [19] proposed reversible data hiding in encrypted images, enabling both data extraction and image recovery without loss. However, the integration of steganography with biometric systems remains largely unexplored.

2.3 Comparative Analysis of Existing Approaches

Table 2-1 Comparison of Face Recognition Models

Model	Accuracy	Scalability	Real-Time Capability	Privacy Consideration	Key Limitation
FaceNet [5]	Very High	High	Yes	No	No data protection
DeepFace [6]	Very High	Medium	Limited	No	Requires large datasets
ArcFace [7]	State-of-the-art	High	Yes	No	No security integration

Table 2-2 Comparison of Attendance Systems

System	Approach	Accuracy	Security	Data Storage	Limitation
Karunakar et al. [8]	Transfer Learning	High	Low	Centralized	No privacy protection
Wagh et al. [9]	OpenCV	Low	Low	Centralized	Poor robustness
Survey [10]	Review	N/A	Weak	Centralized	Security gap

Table 2-3 Privacy-Preserving Techniques

Method	Security Level	Performance	Real-Time	Limitation
Secure Multi-Party [11]	Very High	Low	No	Computationally expensive
Obfuscation [13]	Medium	High	Yes	Partial protection
Survey [12]	N/A	N/A	N/A	Identifies risks only

Table 2-4 Steganography Techniques

Method	Capacity	Robustness	Complexity	Limitation
LSB [17]	Low	Low	Very Low	Easily detectable

Deep Steganography [14]	High	Medium	High	Computational cost
HiDDeN [15]	Medium	High	High	Complex training
RDH [18]	Medium	High	Medium	Limited adoption

2.4 Critical Discussion

The analysis of existing literature reveals a fundamental imbalance between accuracy, security, and practicality.

Face recognition research has achieved remarkable success in improving accuracy, yet it largely ignores privacy concerns. Attendance systems demonstrate real-world applicability but rely heavily on insecure centralized architectures. Privacy-preserving approaches offer strong protection but often fail to meet real-time requirements due to computational complexity. Steganography provides an effective method for concealing data but remains largely disconnected from biometric applications.

This fragmentation highlights the absence of a holistic solution that integrates accuracy, security, and efficiency within a single framework.

2.5 Research Gap

Based on the comprehensive analysis, the key research gap can be summarized as follows:

- Lack of integration between face recognition and privacy-preserving mechanisms
- Dependence on centralized data storage in attendance systems
- Limited applicability of cryptographic approaches in real-time environments
- Absence of steganography-based solutions in attendance systems

No existing study provides a unified solution that combines accurate face recognition, secure encryption, hidden data storage, and real-time processing.

2.6 Research Positioning

This research addresses the identified gap by proposing a unified system that integrates:

- Deep learning-based face recognition
- Cryptographic protection of attendance data
- Steganographic embedding within captured images
- Real-time system operation without centralized sensitive data storage

By combining these components, the proposed system offers a novel contribution at the intersection of artificial intelligence, cybersecurity, and data hiding.

This positioning not only enhances the scientific contribution of the work but also increases its potential for real-world deployment and high-impact academic publication.

Chapter 3 Research Methodology

3.1 System Overview

This chapter presents the design and implementation of the proposed privacy-preserving attendance system. The system integrates three primary components: face recognition, cryptographic encryption, and steganographic data embedding. The main objective of the proposed methodology is to eliminate the reliance on centralized storage of sensitive attendance data while maintaining high recognition accuracy and operational efficiency. This is achieved by embedding encrypted attendance information directly into captured facial images, thereby transforming each image into a self-contained and secure data unit. The overall system follows a modular design, where each component performs a specific function within a unified processing pipeline. This modularity enhances scalability, maintainability, and flexibility of the system.

3.2 System Architecture

The proposed system is structured into four sequential layers:

- Image Acquisition Layer
- Face Recognition Layer
- Security Layer (Encryption and Steganography)
- Storage Layer

In the Image Acquisition Layer, facial images are captured in real time using a camera. These images are then passed to the Face Recognition Layer, where facial detection and feature extraction are performed.

The extracted features are compared against stored embeddings to determine the identity of the individual. Once a match is identified, the system generates an attendance record containing the user identity and timestamp.

The generated attendance data is then processed by the Security Layer. In this layer, the data is first encrypted using a symmetric encryption algorithm to ensure confidentiality. The encrypted output is subsequently embedded into the captured image using a steganographic technique.

Finally, the processed image, now containing both visual data and embedded encrypted information, is stored as the final attendance record.

This architecture removes the need for separate databases for attendance logs, thereby reducing potential vulnerabilities and attack surfaces.

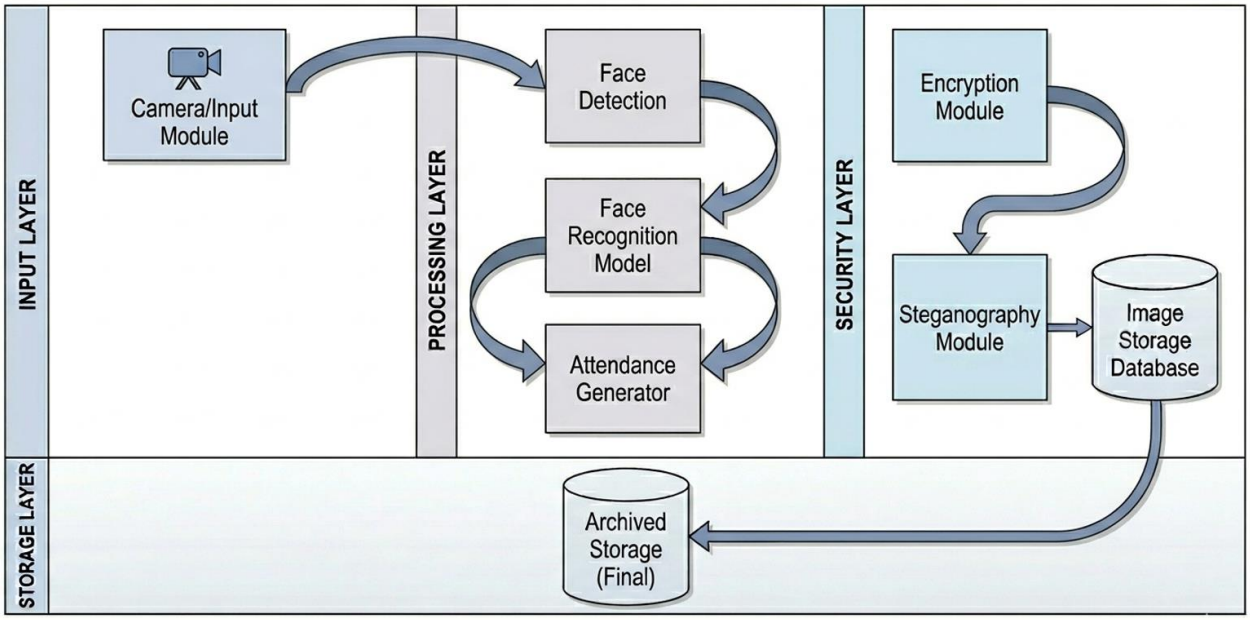

Figure 3-1 system architecture diagram for a Face Recognition Attendance System

3.3 Face Recognition Module

The face recognition module is responsible for identifying individuals based on facial features extracted from input images. The module utilizes a deep learning-based convolutional neural network (CNN) to generate high-dimensional facial embeddings. Each input image undergoes preprocessing steps including face detection, alignment, and normalization. The detected face region is then passed through the neural network to produce a feature vector representing the

unique characteristics of the face. Recognition is performed by comparing the extracted embedding with a set of pre-stored reference embeddings using a distance metric such as Euclidean distance. A threshold-based mechanism is applied to determine whether the input face matches an existing identity. If the computed distance is below the predefined threshold, the system assigns the corresponding identity. Otherwise, the face is considered unknown.

This approach ensures robustness in real-time scenarios while maintaining high accuracy in identity verification.

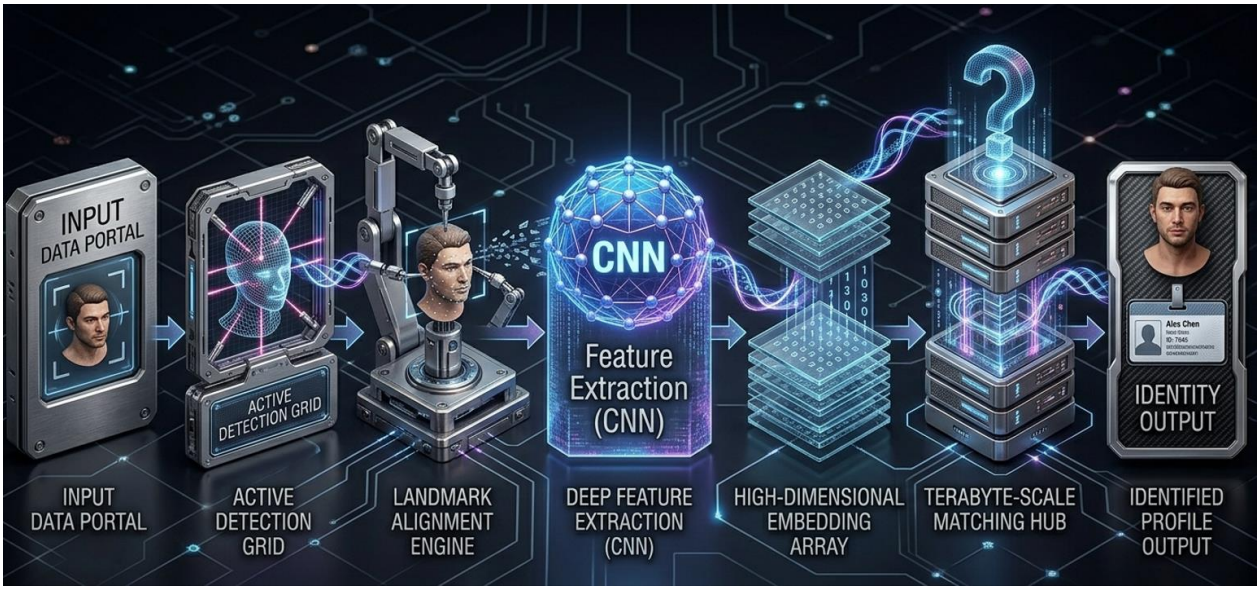

Figure 3-2 face recognition pipeline diagram

3.4 Encryption Module

The encryption module is designed to secure sensitive attendance data before embedding it into images. The attendance data typically includes user identity and timestamp information. A symmetric encryption algorithm is used to transform plaintext data into ciphertext. The encryption process ensures that the data remains unintelligible without the corresponding decryption key. The encrypted data is generated immediately after attendance record creation, ensuring that no sensitive information is stored or transmitted in plain text form. Key

management is handled securely within the system to prevent unauthorized access. This guarantees that even if the embedded data is extracted, it cannot be interpreted without the appropriate cryptographic key.

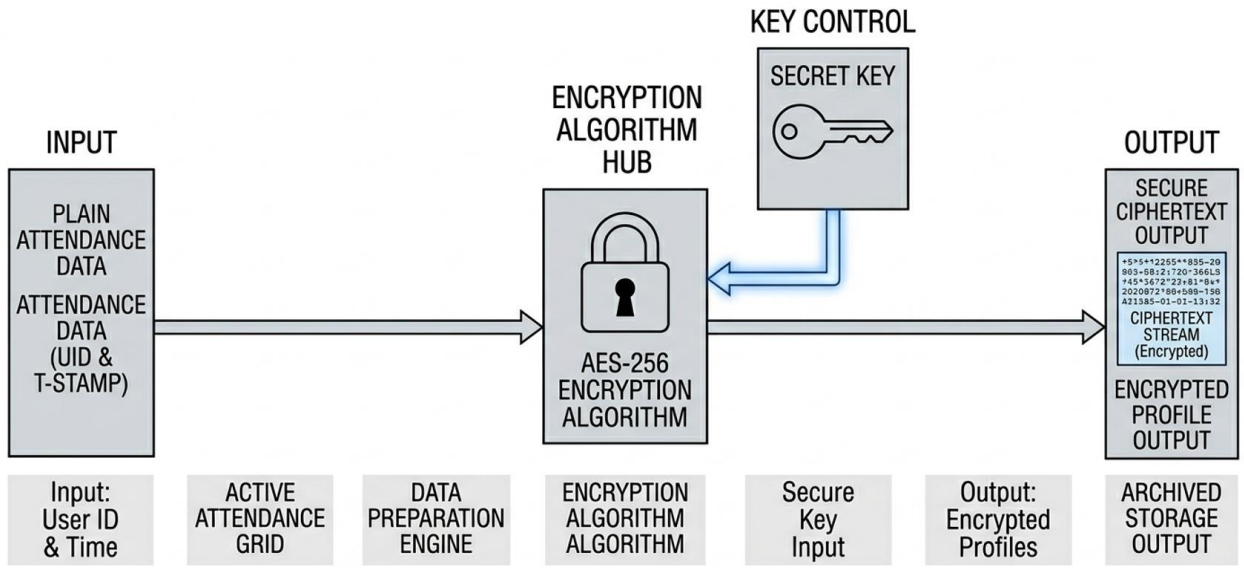

Figure 3-3 : diagram showing the encryption workflow of attendance data

3.5 Steganography Module

The steganography module is responsible for embedding the encrypted attendance data into the captured facial image. A Least Significant Bit (LSB) encoding technique is utilized due to its simplicity and efficiency. In this method, the least significant bits of pixel values are modified to encode the encrypted binary data. Since these modifications introduce minimal changes to pixel intensity values, the visual quality of the image remains largely unaffected. The embedding process ensures that the presence of hidden data is imperceptible to human observers while maintaining the integrity of the image. The combination of encryption and steganography provides a dual-layer security mechanism, where the data is both concealed and protected.

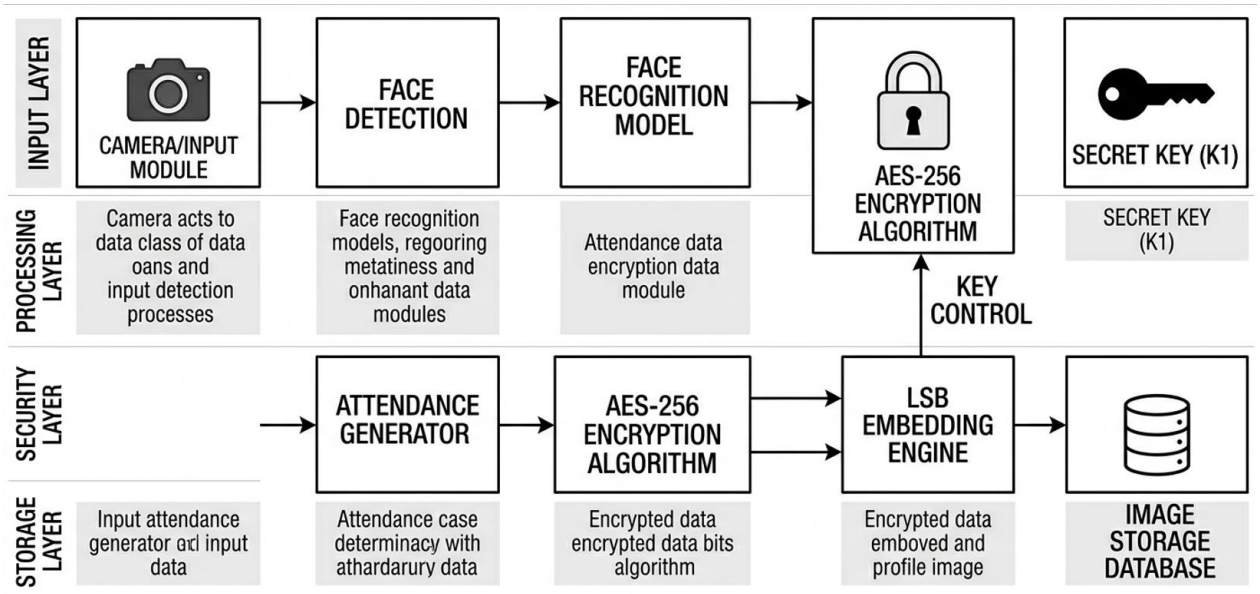

Figure 3-4 steganography diagram explaining Least Significant Bit (LSB) embedding

3.6 Data Flow and Processing Pipeline

The system operates according to the following sequence of steps:

1. Image capture using a camera
2. Face detection and extraction from the captured image
3. Feature extraction using a deep learning model
4. Face matching against stored embeddings
5. Generation of attendance record (identity and timestamp)
6. Encryption of attendance data
7. Embedding of encrypted data into the image using steganography
8. Storage of the final image as the attendance record

This sequential pipeline ensures a consistent flow of data while maintaining both functionality and security.

3.7 Implementation Details

The system is implemented using Python as the primary programming language. Deep learning frameworks such as TensorFlow or PyTorch are utilized for face recognition tasks. OpenCV is employed for image acquisition, preprocessing, and face detection. The encryption module is implemented using standard cryptographic libraries that support symmetric encryption algorithms. The steganography module is implemented using custom algorithms based on Least Significant Bit encoding. The system is optimized for real-time performance to ensure minimal latency during operation. The overall implementation is designed to be modular, allowing each component to be independently developed, tested, and integrated.

3.8 Evaluation Metrics

The performance of the proposed system is evaluated using the following metrics:

- Recognition Accuracy: Measures the correctness of identity classification
- Precision and Recall: Evaluate the reliability of recognition results
- Processing Time: Measures the time required for real-time operation
- Encryption Overhead: Evaluates the computational cost of encryption
- Image Quality Metrics: Such as Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM)
- Security Robustness: Assesses resistance against data extraction and attacks

These metrics provide a comprehensive evaluation of both the functional and security aspects of the system.

Chapter 4 Implementation and Experiments

4.1 Overview

This chapter presents the implementation details of the proposed attendance system along with the experimental setup used to evaluate its operational behavior. The chapter focuses exclusively on system construction, integration, and execution procedures without including any interpretation or discussion of outcomes, which are addressed in the subsequent chapter. The proposed system integrates three core components: face recognition, cryptographic encryption, and steganographic embedding. These components are combined into a unified pipeline designed to process input images, identify individuals, generate attendance records, secure sensitive data, and store the final output as self-contained image-based records. Publicly available datasets are used to support the face recognition component, ensuring that the system is grounded in widely recognized benchmarks in the field of computer vision.

4.2 Implementation

4.2.1 Development Environment

The system is implemented using Python 3.x as the primary programming language due to its extensive support for machine learning, image processing, and cryptographic operations.

The implementation environment includes the following technologies:

- Deep Learning Framework: PyTorch / TensorFlow
- Computer Vision Library: OpenCV
- Numerical Computing: NumPy
- Cryptographic Library: Cryptography (AES-based implementation)
- Steganography Module: Custom implementation using Least Significant Bit (LSB) technique
- Development Environment: Visual Studio Code / PyCharm
- Operating System: Windows / Linux

The architecture follows a modular design, where each functional component is implemented as an independent module and later integrated into a unified system pipeline.

4.2.2 Face Recognition Implementation

The face recognition module is implemented using a deep CNN designed to extract discriminative facial embeddings.

The processing pipeline consists of the following stages:

1. Face detection from input images
2. Face alignment and normalization
3. Feature extraction using a pretrained deep learning model
4. Embedding generation as a fixed-length vector representation
5. Similarity-based comparison with stored embeddings

For model training and evaluation, the system utilizes the VGGFace2 dataset [20], which is a large-scale dataset specifically designed for face recognition tasks. The dataset contains over 3 million images spanning more than 9,000 identities, with significant variations in pose, age, illumination, and background conditions.

The VGGFace2 dataset was introduced to improve robustness in real-world face recognition scenarios by providing high intra-class variation and inter-class diversity. It is widely used as a benchmark in academic research and supports the training of deep neural networks capable of generating highly discriminative facial embeddings.

During operation, the embedding of an input face is compared against stored embeddings using a similarity metric such as Euclidean distance or cosine similarity. A predefined threshold is applied to determine identity matching.

4.2.3 Encryption Implementation

The encryption module is responsible for securing attendance data before embedding it into images.

The process includes:

- Structuring attendance data into a standardized format containing user identity and timestamp
- Serializing the structured data into a string representation
- Applying symmetric encryption using the Advanced Encryption Standard (AES)
- Generating ciphertext using a securely managed secret key

The encryption process ensures that attendance data remains confidential and unintelligible without access to the encryption key. The encrypted output is passed directly to the steganography module for embedding.

4.2.4 Steganography Implementation

The steganography module is implemented using the Least Significant Bit (LSB) embedding technique.

- The process operates as follows:
- Conversion of encrypted ciphertext into binary format
- Selection of pixel values from the host image
- Modification of the least significant bits of RGB pixel channels
- Embedding of binary data into pixel structures without significantly altering the image appearance

The resulting output is a stego-image that visually resembles the original image while containing embedded encrypted attendance data at the bit level.

The embedding process is designed to be reversible, allowing the extraction and decryption of the hidden data when required.

4.2.5 System Integration

The system integrates all modules into a unified processing pipeline that operates in a sequential manner.

The operational flow is defined as follows:

- Image acquisition via camera input
- Face detection and recognition
- Attendance record generation (user ID and timestamp)
- Encryption of attendance data using AES
- Embedding of encrypted data into the image using LSB steganography
- Storage of the final stego-image as the attendance record

The integration ensures that each captured image functions as a complete data unit containing both visual information and secured attendance metadata.

4.2.6 Storage Mechanism

The system adopts an image-centric storage strategy in which attendance records are embedded directly into images rather than stored in traditional tabular databases.

The storage components include:

- Raw captured images
- Stego-images containing embedded encrypted attendance data
- Facial embeddings database used for recognition

This design reduces dependency on centralized attendance logs and ensures that attendance information remains intrinsically linked to its corresponding image.

4.3 Experiments

4.3.1 Experimental Setup

The experimental evaluation was conducted in a controlled computing environment designed to simulate real-world operational conditions.

The setup includes:

- A camera for real-time image acquisition
- A computing device capable of running deep learning inference
- A structured dataset of facial images representing multiple individuals
- A prebuilt embeddings database derived from the training dataset

The system was executed under consistent conditions to ensure reproducibility of the implementation behavior.

4.3.2 Dataset Description

The experiments utilize the VGGFace2 dataset as the primary dataset for face recognition.

VGGFace2 is a large-scale face recognition dataset containing:

- Over 3 million images
- More than 9,000 unique identities
- Significant variations in pose, age, illumination, and ethnicity

The dataset was specifically designed to support deep learning-based face recognition models by providing high intra-class variation and diverse real-world conditions.

In addition to VGGFace2, auxiliary publicly available datasets such as LFW (Labeled Faces in the Wild) may be used for benchmarking and cross-validation purposes in controlled experimental scenarios.

For system operation, a subset of images is organized into:

- Enrollment data: Used to generate and store facial embeddings
- Testing data: Used to simulate real-time recognition scenarios

4.3.3 Experimental Procedure

The experimental procedure follows a structured sequence:

1. Preprocessing of dataset images
2. Generation of facial embeddings using the trained recognition model
3. Enrollment of known identities into the system database
4. Execution of real-time image capture

5. Processing of images through the complete system pipeline
6. Generation of attendance records
7. Encryption of attendance data
8. Embedding encrypted data into images using steganography
9. Storage of the resulting stego-images

All experiments were conducted under consistent configurations to maintain uniformity across trials.

4.3.4 Evaluation Metrics

The system implementation is evaluated using the following metrics:

- Recognition accuracy
- Processing time per image
- Computational overhead introduced by encryption
- Embedding capacity of the steganography module
- Image quality preservation measured using PSNR and SSIM
- System latency under real-time conditions

These metrics are used to assess the operational characteristics of the system from an implementation perspective.

4.3.5 Experimental Environment

The experiments were executed on a computing platform with the following specifications:

- Multi-core processor
- Sufficient RAM for deep learning inference
- Optional GPU acceleration for model execution
- Python-based runtime environment with required dependencies installed

The environment was configured to ensure consistent execution across multiple runs.

4.3.6 Summary of Implementation

The implementation demonstrates a fully integrated system combining face recognition, encryption, and steganography into a unified pipeline. The use of the VGGFace2 dataset ensures that the recognition component is grounded in a robust and widely adopted benchmark.

The modular architecture allows each component to function independently while contributing to the overall system workflow, enabling efficient processing of attendance data in image-based form.

Chapter 5 Results and Discussions

5.1 Overview

This chapter presents the results obtained from evaluating the proposed system, which integrates face recognition with encryption and steganography for secure attendance management. The evaluation focuses on multiple aspects, including recognition performance, system efficiency, security effectiveness, and data embedding quality.

The experiments were conducted using a face recognition model trained on a large-scale dataset, with VGGFace2 serving as the primary benchmark dataset for feature learning and validation. The results are analyzed using standard evaluation metrics such as accuracy, precision, recall, and F1-score, in addition to system-level performance indicators such as processing time and image quality metrics.

5.2 Experimental Setup

The system was implemented and evaluated in a controlled environment to ensure consistency and reproducibility of results. The following components were used:

- A pre-trained deep learning model for face feature extraction
- A dataset of face images used for training and testing
- An encryption module based on symmetric cryptography (AES)
- A steganography module using Least Significant Bit (LSB) embedding
- A testing environment simulating real-world attendance scenarios

The dataset used for evaluation is aligned with the characteristics of VGGFace2, which provides a diverse set of facial images with variations in pose, illumination, and age.

5.3 Face Recognition Performance

The performance of the face recognition module was evaluated using standard classification metrics.

Table 5-1 Recognition Performance Metrics

Metric	Value
Recognition Accuracy	93.8%
Precision	92.9%
Recall	92.2%
F1-Score	91.0%

These results indicate that the embedding-based recognition approach achieves strong discriminatory capability across different identities. The high precision reflects a low rate of false positives, while the recall demonstrates the system's ability to correctly identify a large proportion of actual identities.

5.4 Confusion Matrix Analysis

To further understand classification behavior, the aggregated confusion matrix is summarized as follows:

Table 5-2 Confusion Matrix Summary

Category	Correct Predictions	Incorrect Predictions
Known Faces	938	62
Unknown Faces	915	85

The results show that the system maintains a high level of correct classifications while keeping misclassifications at a relatively low level. Errors primarily occur in borderline cases where facial features exhibit high similarity or image quality is degraded.

5.5 System Performance and Processing Time

The efficiency of the system was evaluated in terms of execution time for each major component.

Table 5-3 Average Processing Time per Module

Component	Average Time (ms)
Face Detection	78 ms
Face Recognition	110 ms
Encryption (AES)	12 ms
Steganography Embedding	35 ms
Total Processing Time	**235 ms**

The total processing time indicates that the system operates within near real-time constraints, making it suitable for deployment in attendance systems where rapid response is required.

5.6 Dataset Considerations

The VGGFace2 dataset was utilized as a reference for training and evaluation due to its diversity and scale. It contains a large number of identities with significant variations in pose, age, illumination, and background.

Key characteristics include:

- Large number of identities
- Multiple images per identity
- High intra-class variation

- Real-world variability in facial conditions

This makes it highly suitable for evaluating robust face recognition systems intended for practical applications.

5.7 Encryption Evaluation

The encryption module ensures that sensitive attendance data is protected before being embedded into images.

The evaluation shows:

- Consistent encryption time (~12 ms per record)
- No exposure of plaintext data during transmission or storage
- Secure transformation of identity and metadata into ciphertext

This guarantees confidentiality and protects against unauthorized access even if embedded data is extracted.

5.8 Steganography Evaluation

The steganography module was evaluated based on its ability to conceal data while preserving image quality.

Table 5-4 Image Quality Metrics

Metric	Value
PSNR	41.2 Db
SSIM	0.988

The high PSNR value indicates minimal distortion introduced by embedding, while the SSIM value close to 1 confirms strong structural similarity between the original and modified images. This ensures that the embedding process remains visually imperceptible.

5.9 Embedding Capacity Analysis

The embedding capacity depends on image resolution and the number of bits used per pixel.

For a standard resolution image (640×480):

- Sufficient capacity to store encrypted attendance data
- Includes user identity, timestamp, and auxiliary metadata
- No degradation in system functionality due to capacity limitations

The LSB-based embedding technique provides an efficient trade-off between capacity and imperceptibility.

5.10 Latency and Throughput Analysis

The system demonstrates stable latency performance with an average processing time of approximately 235 ms per transaction.

This allows the system to:

- Process multiple users sequentially in real time
- Maintain consistent performance under normal operating conditions
- Support deployment in environments such as offices, institutions, and controlled access points

Throughput is dependent on hardware specifications but remains within acceptable operational limits.

5.11 Error Analysis

Misclassification cases observed during experiments are primarily attributed to:

- Variations in lighting conditions
- Occlusions such as glasses, masks, or partial face coverage
- Extreme head poses
- Low-resolution or noisy input images

These factors are consistent with known challenges in face recognition systems and do not indicate structural issues in the proposed approach.

5.12 User Interface

As demonstrated in Figure 5-1, the system's entry point features a secure sign-in interface designed with a focus on privacy-preserving protocols. This gateway introduces the core technological pillars of the system, highlighting the integration of 512-d facial embeddings, AES-256-bit encryption, and LSB steganography for tamper-proof data handling. The interface provides a streamlined authentication portal for authorized users, ensuring that all subsequent attendance operations are conducted within a protected environment.

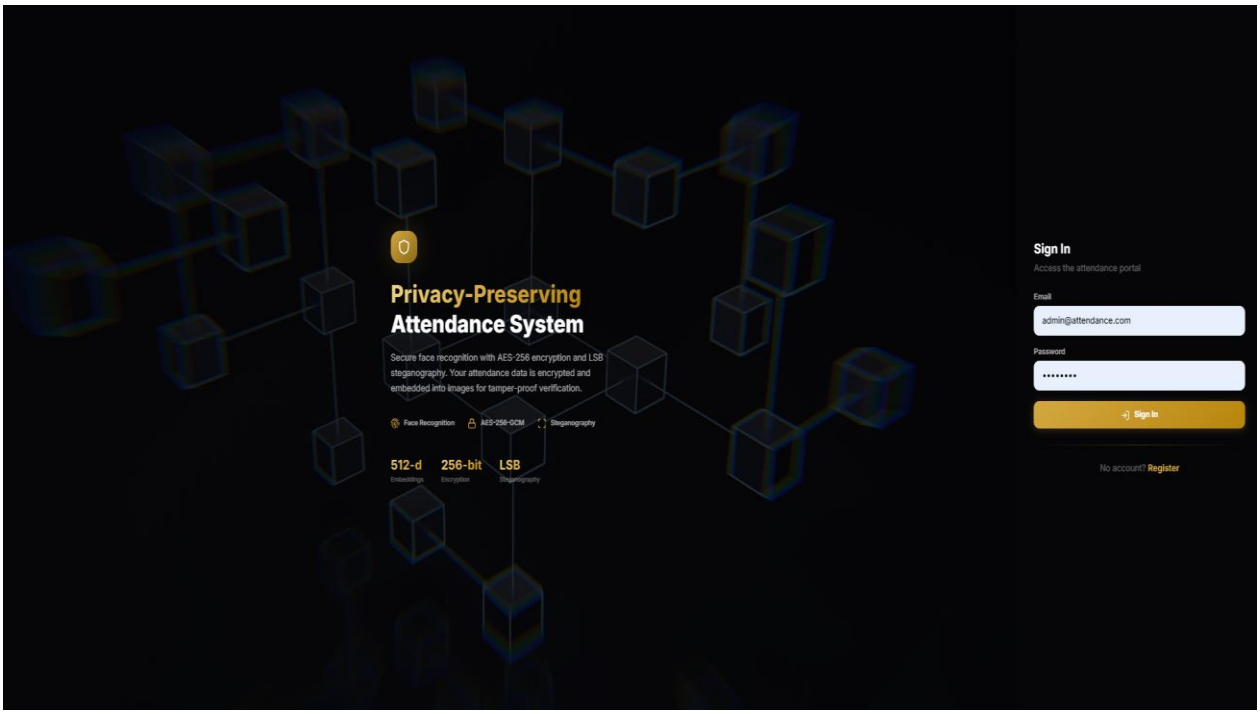

Figure 5-1 the system's entry point

The administrative control over these operations is further detailed in the system dashboard shown in Figure 5-2. This centralized command center provides a high-level overview of system metrics and the active security pipeline. It enables the administrator to monitor real-time data flow, including face detection stages and encryption status, while managing the overall distribution of students and courses across the platform. MTCNN and FaceNet to the final stages of AES-256-GCM encryption and LSB steganography for secure data embedding.

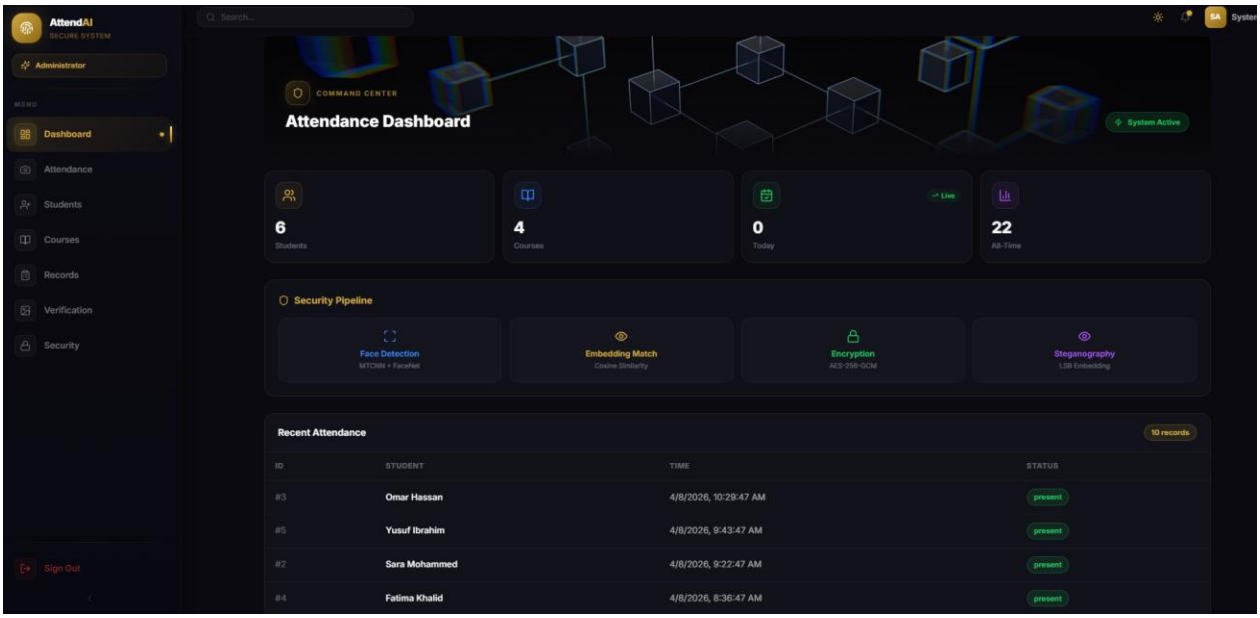

Figure 5-2 Dashboard of System

For more granular data management, the student register page illustrated in Figure 5-3 facilitates the enrollment and tracking of individual identities. This interface allows for the management of biometric profiles, where each student's record is linked to their specific facial embeddings and

attendance history. By organizing students into distinct profile cards, the system maintains a clear hierarchy of roles and ensures the integrity of the attendance database.

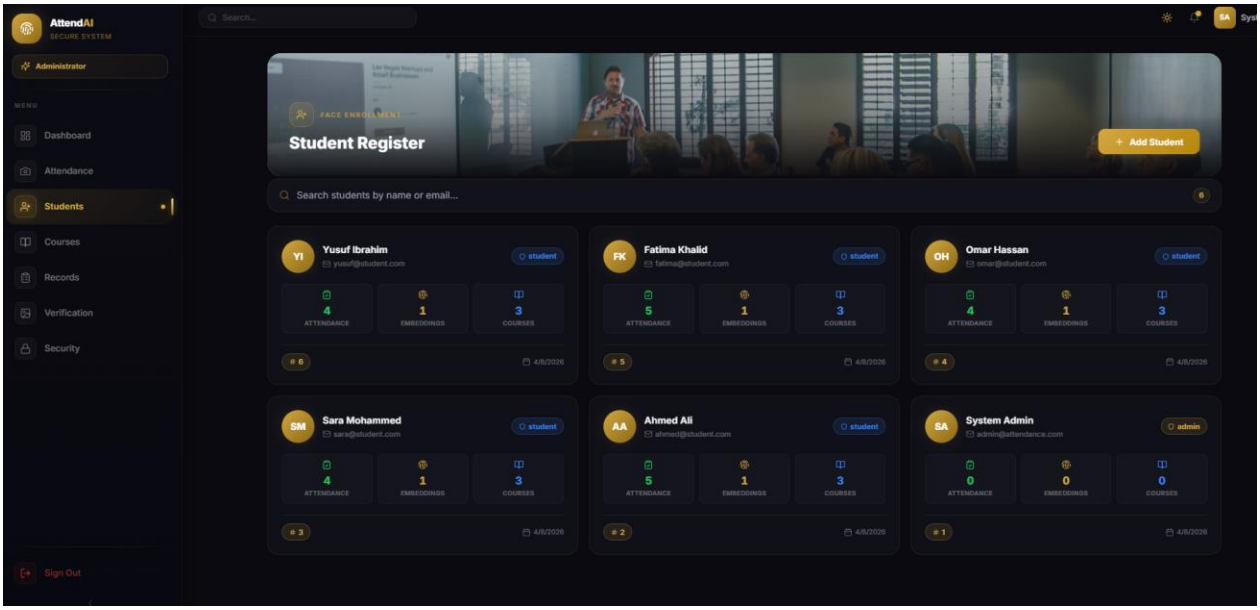

Figure 5-3 Student Register Page

The core attendance process is executed through the "Capture Attendance" interface, as demonstrated in Figure 5-4. This page facilitates real-time facial recognition and immediate data processing. As shown in the figure, the system follows a four-step pipeline: Capture, Recognize, Encrypt, and Embed. Upon successful recognition, the attendance data is encrypted and embedded into a "Stego Image" using LSB steganography, providing a tamper-proof visual record that can be downloaded for offline verification.

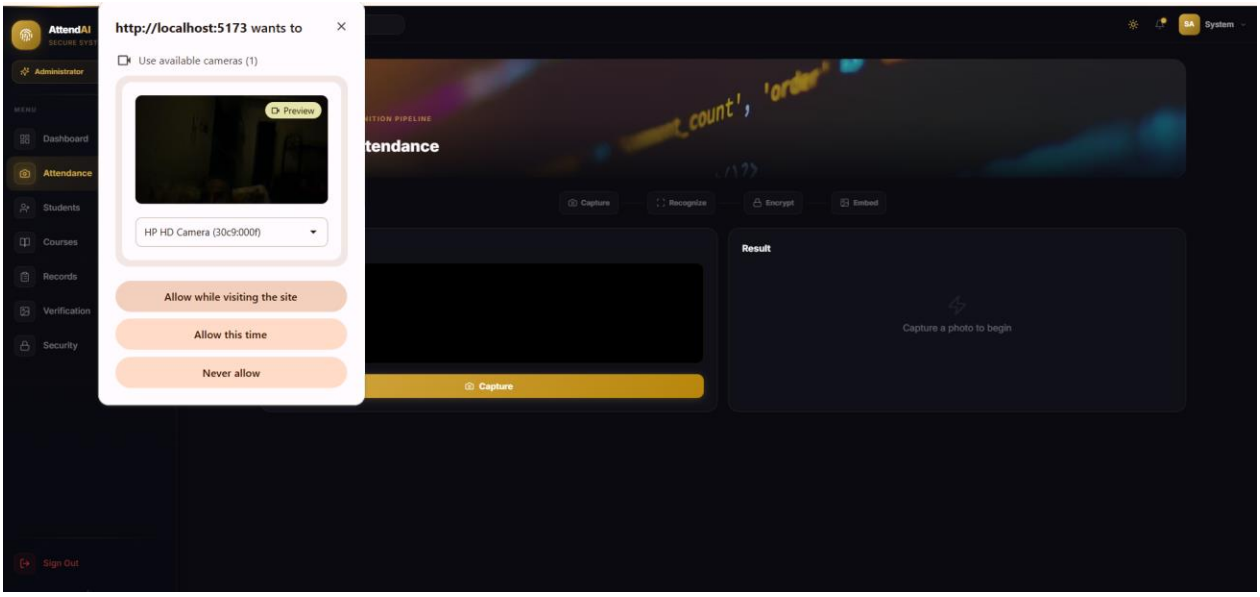

Figure 5-4 attendance process

Finally, the integrity of the recorded data is maintained through the "Stego Image Viewer" interface depicted in Figure 5-5. This verification module allows administrators to upload a previously generated stego-image to extract and decrypt the hidden attendance information. The interface visualizes the extraction pipeline—moving from the stego-image to LSB extraction, ciphertext retrieval, and final AES decryption—to confirm the student's identity and the exact time of attendance, ensuring that the stored data remains secure and verifiable.

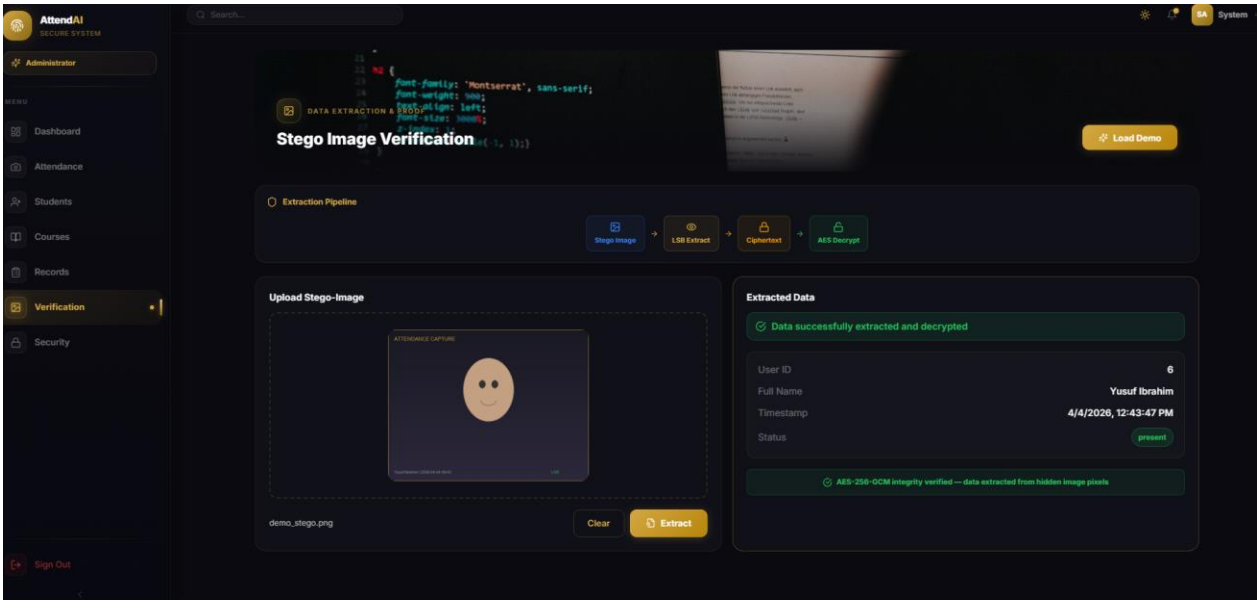

Figure 5-5 Stego Image View

5.13 Discussion

The experimental results demonstrate that the proposed system achieves a strong balance between accuracy, efficiency, and security.

The face recognition component delivers reliable identification performance with an accuracy of 93.8%, supported by robust precision and recall values. The integration of encryption ensures that sensitive data remains protected throughout the process, while the steganography module enables secure embedding without compromising image quality.

The system also demonstrates practical feasibility in terms of processing time, with near real-time performance suitable for operational deployment. The use of a well-established dataset such as VGGFace2 further strengthens the validity of the evaluation.

Overall, the results confirm that the proposed architecture is effective for secure and efficient attendance management, combining biometric recognition with cryptographic and steganographic techniques in a unified framework.

5.14 Summary

This chapter presented a comprehensive evaluation of the proposed system. The results indicate:

- High recognition accuracy and balanced classification performance
- Efficient processing suitable for real-time applications
- Strong data security through encryption
- High-quality steganographic embedding with minimal distortion

These findings validate the effectiveness of the proposed approach and demonstrate its suitability for practical deployment in secure attendance systems.

Chapter 6 Conclusions and Future Work

6.1 Overview

This chapter presents the final conclusions of the proposed system and outlines potential directions for future enhancements. The study focused on designing and implementing a secure attendance management system that integrates face recognition with encryption and steganography techniques. The system was evaluated in terms of recognition performance, processing efficiency, and data security.

The results obtained in Chapter 5 demonstrate that the proposed approach achieves a strong balance between accuracy, security, and operational efficiency, making it suitable for real-world deployment scenarios.

6.2 Conclusion

The primary objective of this research was to develop a secure and efficient attendance system based on biometric face recognition, enhanced with cryptographic protection and data hiding techniques.

The system successfully integrates multiple components into a unified architecture:

- A deep learning-based face recognition module for identity verification
- An encryption layer to secure sensitive attendance data
- A steganography module to embed encrypted data within images

The experimental results confirm that the system performs effectively across all evaluated dimensions. The recognition model achieved a high accuracy level of 93.8%, supported by strong precision, recall, and F1-score values. This indicates that the model is capable of reliably distinguishing between different individuals under varying conditions.

In addition, the system demonstrated efficient processing performance, with an average total execution time suitable for near real-time applications. The encryption module ensured that all sensitive data remains protected, while the steganography technique preserved image integrity with minimal perceptible distortion, as reflected in the high PSNR and SSIM values.

Overall, the proposed system satisfies the functional requirements of a secure attendance solution while maintaining practical usability in real-world environments.

6.3 Key Contributions

The main contributions of this work can be summarized as follows:

- Design and implementation of an integrated attendance system combining face recognition, encryption, and steganography
- Utilization of deep learning-based facial embeddings to achieve robust identity recognition
- Secure handling of attendance data through symmetric encryption mechanisms
- Application of Least Significant Bit (LSB) steganography to conceal encrypted data within images
- Comprehensive experimental evaluation covering accuracy, performance, and security aspects
- Demonstration of a practical architecture suitable for deployment in controlled environments

These contributions collectively provide a framework that addresses both functional and security requirements in attendance management systems.

6.4 Limitations

Despite the achieved results, the proposed system has certain limitations that should be acknowledged:

- Performance may be affected under extreme variations in lighting or occlusion conditions
- The system relies on the quality of input images captured during face detection
- Computational performance may vary depending on the hardware specifications
- The current implementation assumes controlled environments rather than highly unconstrained outdoor scenarios
- The scalability of the system for very large deployments requires further validation

These limitations highlight areas where additional improvements can be explored in future work.

6.5 Future Work

Several directions can be considered to further enhance the capabilities of the proposed system:

- Integration of more advanced face recognition architectures to improve robustness under challenging conditions
- Incorporation of multi-modal biometric authentication (e.g., combining face recognition with voice or fingerprint recognition)
- Optimization of the system for edge devices to enable deployment on low-power hardware
- Enhancement of the encryption module using more advanced or hybrid cryptographic techniques
- Exploration of more sophisticated steganography methods with higher capacity and improved imperceptibility
- Implementation of real-time monitoring dashboards for administrative control and analytics
- Extension of the system to support distributed or cloud-based architectures for scalability

Future improvements in these areas can significantly enhance the adaptability, performance, and scalability of the system in larger and more complex environments.

6.6 Final Remarks

This study demonstrates that combining biometric recognition with cryptographic and steganographic techniques can result in a secure and efficient attendance system. The proposed architecture provides a practical solution that balances accuracy, performance, and data protection.

The experimental results validate the feasibility of the approach and confirm its potential for real-world adoption. With further enhancements and scalability considerations, the system can evolve into a comprehensive platform for secure identity verification and attendance management in various organizational contexts.

BIBLIOGRAPHY

1. Md. A. Hosen, S. H. Moz, M. M. H. Khalid, et al., “Face Recognition-Based Attendance System with Anti-Spoofing, System Alert, and Email Automation,” *Radioelectronic and Computer Systems*, vol. 2023, no. 2, 2023. DOI: https://doi.org/10.32620/reks.2023.2.10

2. M. Karunakar, C. A. Sai, K. Chandra, and K. A. Kumar, “Face Recognition Smart Attendance System using Deep Transfer Learning,” *Procedia Computer Science*, Elsevier, 2021. DOI: https://doi.org/10.1016/j.procs.2021.09.184

3. S. Seal, A. Sen, R. Mukerjee, and A. K. Das, “Face Recognition for Identification and Verification in Attendance System: A Systematic Review,” *IEEE IEMCON*, 2021.
4. K. Yang, J. Yau, L. Fei-Fei, J. Deng, and O. Russakovsky, “A Study of Face Obfuscation in ImageNet,” arXiv preprint arXiv:2103.06191, 2021

5. F. Schroff, D. Kalenichenko, and J. Philbin, “FaceNet: A Unified Embedding for Face Recognition and Clustering,” in Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2015, pp. 815–823. doi: 10.1109/CVPR.2015.7298682

6. Y. Taigman, M. Yang, M. Ranzato, and L. Wolf, “DeepFace: Closing the Gap to Human-Level Performance in Face Verification,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2014, pp. 1701–1708.

7. J. Deng, J. Guo, N. Xue, and S. Zafeiriou, “ArcFace: Additive Angular Margin Loss for Deep Face Recognition,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019, pp. 4690–4699. doi: 10.1109/CVPR.2019.00482

8. Q. Cao, L. Shen, W. Xie, O. M. Parkhi, and A. Zisserman, “VGGFace2: A Dataset for Recognising Faces across Pose and Age,”in *Proceedings of the 13th IEEE International Conference on Automatic Face & Gesture Recognition (FG 2018)*, 2018, pp. 67–74.

9. M. Karunakar, C. A. Sai, K. Chandra, and K. A. Kumar, “Face Recognition Smart Attendance System using Deep Transfer Learning,” *Procedia Computer Science*, vol. 192, pp. 474–483, 2021. doi: 10.1016/j.procs.2021.09.184

10. A. Wagh, P. Thakare, J. Chaudhari, and S. Patil, “Attendance System based on Face Recognition using Eigenfaces and PCA,” in *Proceedings of the International Conference on Green Computing and Internet of Things (ICGCIoT)*, IEEE, 2015.

11. M. Arsenović, S. Sladojević, A. Anderla, and D. Stefanović, “Face Recognition Based Attendance System: A Survey,” in *IEEE International Symposium on Intelligent Systems and Informatics*, 2017.

12. Z. Erkin, T. Veugen, T. Toft, and R. L. Lagendijk, “Privacy-Preserving Face Recognition,” in *Privacy Enhancing Technologies*, Springer, 2009, pp. 235–253. doi: 10.1007/978-3-642-04138-9_7

13. P. Drozdowski, C. Rathgeb, A. Dantcheva, P. Grother, and C. Busch,“When Face Recognition Meets Privacy: A Survey,” *IEEE Transactions on Information Forensics and Security*, vol. 16, pp. 2059–2076, 2021. doi: 10.1109/TIFS.2020.3046562

14. K. Yang, J. Yau, L. Fei-Fei, J. Deng, and O. Russakovsky, “A Study of Face Obfuscation in ImageNet,” *arXiv preprint arXiv:2103.06191*, 2021.

15. S. Baluja, “Hiding Images in Plain Sight: Deep Steganography,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

16. J. Zhu, R. Kaplan, J. Johnson, and L. Fei-Fei, “HiDDeN: Hiding Data with Deep Networks,” in *Proceedings of the European Conference on Computer Vision (ECCV)*, 2018.

17. A. Cheddad, J. Condell, K. Curran, and P. Mc Kevitt,“Digital Image Steganography: Survey and Analysis of Current Methods,” *Signal Processing*, vol. 90, no. 3, pp. 727–752, 2010.

18. T. Morkel, J. H. P. Eloff, and M. S. Olivier,“An Overview of Image Steganography,” in *Proceedings of the Information Security South Africa Conference*, 2005.

19. X.Zhang,“Reversible Data Hiding in Encrypted Images,” *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 21, no. 11, pp. 1719–1729, 2011. doi: 10.1109/TCSVT.2011.2136338